GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 4

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Recap of Previous Lecture

Topic

Topic

Topic

Topic

Topic

Question

A only if B $\Rightarrow$ $A \to B$

A if B $\Rightarrow$ $B \to A$

Turn on Slide map

Topics to be Covered

Topic PYQ

Topic Statements $A \to B$, $B \to A$

Topic Statement $A \equiv B$

Topic

Topic

DISCIPLINE
IS DOING
WHAT NEEDS
TO BE DONE,
EVEN IF
YOU DON'T
WANT TO
DO IT.
PICTUREQUOTES.COM

[NAT]

* imp.

Following statements are equivalent,

- Argument (P1, P2, P3,.....,.Pn} yields Q is valid. $(P_1, P_2, P_3 - - - P_N) \mapsto Q$
- {P1, P2, P3,....., Pn} Logically implies Q is valid
- {P1 $\wedge$ P2 $\wedge$ P3 $\wedge$..... A $\wedge$ Pn} →Q is a tautology

[MCQ]

Which of the following is/are false

done

A $a \rightarrow (b \vee c) \equiv (a \wedge \sim b) \rightarrow c$

B $(P \rightarrow Q) \wedge (R \rightarrow Q) \equiv (P \vee R) \rightarrow Q$

C $(P \rightarrow Q) \vee (R \rightarrow Q) \equiv (p \wedge R) \rightarrow Q$

[MCQ]

Which of the following arguments is/are valid

done

A. Conclusion R follows prom the premises $\{P \rightarrow (Q \rightarrow R), P \wedge Q\}$

B. $\{P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P\}$ logically implies 'S'

C. $(P \wedge Q)$ follows from the premises $\{(P \vee Q), \sim P\}$

D. $\{\sim T \rightarrow \sim R, \sim S, T \rightarrow W, R \vee S\} \mapsto W$

[NAT]

Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ yields $\sim P$

done

Proof by Contradiction

(Skip this)

→ another method to prove that argument is valid/invalid

→ argument $\underbrace{(P_1, P_2 - - - P_N) \vdash Q}_{\text{Is this valid or not}}$

1) we assume that this is invalid i.e we assume Q is false OR $\neg Q$ is true.

2) Now we put $[P_1, P_2, P_3 - - - P_N, \neg Q]$

Now take $\neg Q$ = True, now try to make all the Premises = True

If this Condition exist then argument is

→ Invalid

→ Else Valid

[NAT]

Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ yields $\sim P$

done

Proof by Contradiction

(done)

Conditional Proof

Rule ⇒ $\underbrace{(P_1, P_2, \ldots P_N)}_{\text{Premise}}$ logically implies $\underbrace{(Q \rightarrow R)}_{\text{Conclusion}}$

This is equivalent to

$(P_1, P_2, P_3 \ldots P_N, Q)$ logically implies (R)

[NAT]

Check whether the argument is valid or invalid

Valid

$\{P \to (Q \to S), \sim R \vee P, Q\}$ logically implies $(R \to S)$

$P \to (Q \to S)$ (1, 0, 1, 0)

P	Q	R	S
1	1	1	0
1	1	0	0

(12,14) false

$\sim R \vee P$ (1, 0)

P	Q	R	S
0	0	1	0
0	0	1	1
0	1	1	0
0	1	1	1

$\Rightarrow$ 2,3,6,7

$Q = 0$

false

P	Q	R	S
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1

0,1,2,3, 8,9,10,11

~~0~~ ~~1~~ ~~2~~ ~~3~~ 4 5 ~~6~~ ~~7~~ ~~8~~

~~9~~ ~~10~~ ~~11~~ ~~12~~ 13 ~~14~~ 15

$R \to S$ (1, 0)

P	Q	R	S
0	0	1	0
0	1	1	0
1	0	1	0
1	1	1	0

(2,6,10,14) False

T: 0,1,3,4,5,7,8,9,11,12,13,15

Example 1:

$$\{P \vee Q, \neg P\} \text{ yields } Q$$

Check whether the following argument is valid or invalid.

$(P \vee Q, \neg P)$ yield Q

True only if Q = True bcoz P = False

True (P = False)

Example 2: Prove valid

$$\{P \to R, Q \to R, \neg R\} \text{ yields } \neg P \wedge \neg Q$$

Check whether the following argument is valid or invalid.

$P \to R$, $Q \to R$, $\neg R$

P	Q	R
1	0	0
1	1	0

(4,6)

P	Q	R
0	1	0
1	1	0

(2,6)

R=1

P	Q	R
0	0	1
0	1	1
1	0	1
1	1	1

(1,3,5,7)

0 ~~1~~ ~~2~~ ~~3~~

~~4~~ ~~5~~ ~~6~~ ~~7~~

PQR = 0,0,0

$\neg P \wedge \neg Q$

$1 \wedge 1 = 1$

Example 3:

$$\boxed{\{P \to Q, Q \to R\} \text{ yields } P \to R}$$

Check whether the following argument is valid or invalid.

Property.

Example 4:

$$\{P \to (Q \wedge R), \neg R\} \text{ yields } \neg P$$

Check whether the following argument is valid or invalid.

H.W

Example 5:

$$\{\neg(P \vee Q), P\} \text{ yields } \neg Q$$

Check whether the following argument is valid or invalid.

H.W

Some Important Statements

1. P only if Q vs P if Q

$(P \to Q)$ $(Q \to P)$ ✓

Q is necessary Condition of P

Q is sufficient Condition of P

P " " " " Q

P " necessary " " Q $\Rightarrow$ Q only if P

$Q \to P$

$(Q \text{ if } P \Rightarrow P \to Q)$

Some Important Statements

1. Q is necessary condition for P vs Q is sufficient condition for P

$\Downarrow$

P only if Q

$\Downarrow$

$P \to Q$

$\Downarrow$

P if Q

$\Downarrow$

$Q \to P$

- Q is necessary Condition for P ⇒ $\boxed{\text{P only if Q}}$ $\boxed{P \to Q}$
- Q is sufficient " " P ⇒ P if Q $Q \to P$
- P " " " " Q ⇒ Q if P $P \to Q$
- P " necessary " " " ⇒ Q only if P $Q \to P$

Some Important Statements

A

(If Sooraj is Present then Rahul must be Present)

B

B if A

Comparison: Q if P

$P \to Q$

$P \to Q$

If P happens, Q must happen. → Sufficient

$\Leftarrow$ For P to happen, Q must already be true. → necessary.

necessary Condition

Q is necessary for P

A

For Sooraj to be Present Rahul shd already be Present.

B

A only if B

Some Important Statements

A implies B

1. If A, then B

$A \to B$

2. A is sufficient for B $\Rightarrow$ B if A $\Rightarrow (A \to B)$

↳ B तब aega

Jab A hoga

3. B is necessary for A

A only if B $\Rightarrow (A \to B)$

4. A only if B $A \to B$

5. Not A or B ($\neg A \vee B$)

ical

Some Important Statements

A implies B

6. **Contrapositive: If not B, then not A** (¬B → ¬A)

$(\neg B \to \neg A)$ ✓

1

7. **B follows from A**

~~B only if A~~

B if A

$(A \to B)$

8. **A implies B** in symbolic form: $A \to B$

9. **B is true whenever A is true** $\Rightarrow$ B if A $\Rightarrow (A \to B)$

B if A

~~B only if A~~

10. **A guarantees B**

$A \to B$

[NAT]

1. $P \to Q \equiv P \to Q$
2. P implies $Q \equiv P \to Q$ ✓
3. If P then $Q \equiv P \to Q$ ✓
4. P only if $Q \equiv P \to Q$ ✓
5. P is sufficient condition for $Q \equiv$ Q if P $\Rightarrow P \to Q$
6. Q is necessary condition for $P \equiv$ P only if Q $\Rightarrow P \to Q$

[NAT]

7. P if Q ≡ if Q then P

8. P when Q ≡ ~~Only if~~ , P if Q ⇒ Q → P

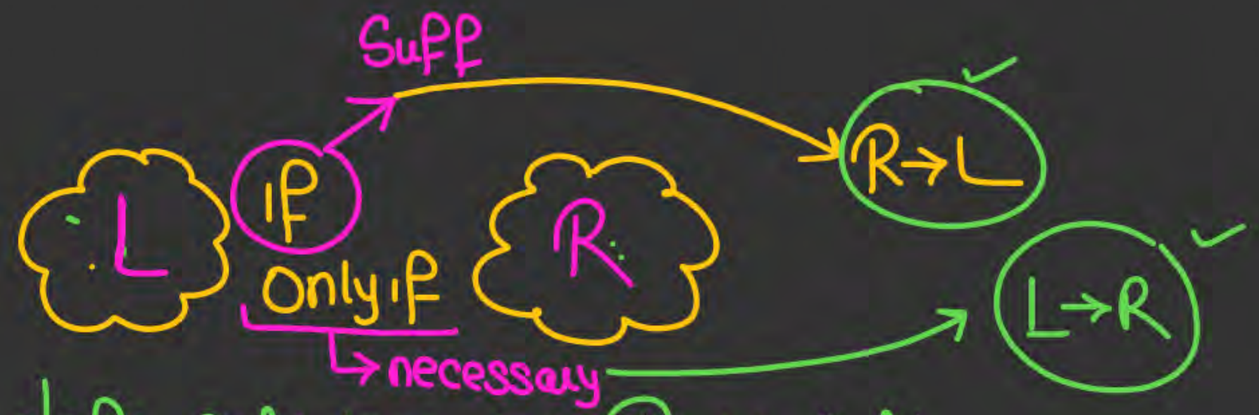
9. P follows from Q ≡ P if Q ⇒ Q → P

10. P unless Q ≡ , P only if Q

P → Q

(Rahul is present unless Sooraj is present)

Jabtak



left wali depend on Right wali

A is true if B is true A if B $B \to A$

A is true only if B is true A only if B $A \to B$

A whenever B $\Rightarrow$ A if B $\Rightarrow$ $B \to A$.

Some Important Statements

A equivalent to B $(A \equiv B)$

1. A if and only if B

A if B, A only if B

$B \to A$ $A \to B$

$A \leftrightarrow B \Rightarrow A \equiv B$

2. A is equivalent to B

$A \equiv B$

3. A is true if and only if B is true

$A \to B$, $B \to A$ $\Rightarrow A \leftrightarrow B$

4. A if B and A only if B

$B \to A$ $A \to B$ $A \leftrightarrow B$

Some Important Statements

A equivalent to B

5. **A implies B and B implies A**

$A \Leftrightarrow B$
$B \Leftrightarrow A$ $\rightarrow A \equiv B$

6. **A is necessary and sufficient for B**

$A \rightarrow B$
$B \rightarrow A$ $\rightarrow A \equiv B$

7. **A whenever B, and A only if B**

$B \rightarrow A$
$A \rightarrow B$ $\rightarrow A \equiv B$

8. **A and B have the same truth value**

$A \equiv B$

e false.

Some Important Statements

A equivalent to B

9. **Both A and B are true, or both are false**

This means they have Same Truth table
$A \equiv B$, $A \leftrightarrow B$

10. **A is true exactly when B is true**

$(A \rightarrow B)$

11. **Not A if and only if not B** ($\neg A \leftrightarrow \neg B$)

$(\neg A \Leftrightarrow \neg B)$

A is true when B is true $\longrightarrow$ $(A \text{ if } B) \Rightarrow (B \rightarrow A)$ ✓

$\Rightarrow$ A is true only when B is true $\longrightarrow$

A is true exactly when B is true $\longrightarrow$

Same

A only if B $\Rightarrow (A \rightarrow B)$

[MCQ]

Which of the following implication is/are valid

or

which of the following is/are a tautology

Q.

(A) $(P \wedge Q) \rightarrow (P \leftrightarrow Q)$

(B) $(P \leftrightarrow Q) \rightarrow (P \rightarrow Q)$

(C) $(P \leftrightarrow Q) \rightarrow (P \rightarrow \sim Q)$

(D) $Q \rightarrow (P \rightarrow Q)$

[MCQ]

Which of the following is/are false

Q

A $a \rightarrow (b \vee c) \equiv (a \wedge \sim b) \rightarrow c$

B $(P \rightarrow Q) \wedge (R \rightarrow Q) \equiv (P \vee R) \rightarrow Q$

C $(P \rightarrow Q) \vee (R \rightarrow Q) \equiv (p \wedge R) \rightarrow Q$

[MCQ]

Which of the following arguments is/are valid

Q

A. Conclusion R follows prom the premises $\{P \rightarrow (Q \rightarrow R), P \wedge Q\}$

B. $\{P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P\}$ logically implies 'S'

C. $(P \wedge Q)$ follows from the premises $\{(P \vee Q), \sim P\}$

D. $\{\sim T \rightarrow \sim R, \sim S, T \rightarrow W, R \vee S\} \mapsto W$

[NAT]

Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ yields $\sim P$

Proof by Contradiction

[NAT]

Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ yields $\sim P$

Proof by Contradiction

Converse, Inverse & Contrapositive in Propositional Logic

- In propositional logic, converse, inverse, and contrapositive are terms related to implication statements (of the form $P \rightarrow Q$P→Q, meaning "if PP, then QQ"). These variations help in understanding the relationships between statements and their logical equivalents.

Converse, Inverse & Contrapositive in Propositional Logic

- 1. Converse: The converse of an implication $P \rightarrow Q$ is the statement $Q \rightarrow P$.
- It switches the hypothesis and conclusion.
- Original: $P \rightarrow Q$ "If P, then Q."
- Converse: $Q \rightarrow P$ "If Q, then P."
- Note that the converse is not necessarily logically equivalent to the original implication.
- Example: Original: "If it rains, then the ground is wet.
- "Converse: "If the ground is wet, then it rains." (The ground could be wet for other reasons, so this is not logically equivalent.)

Converse, Inverse & Contrapositive in Propositional Logic

- Inverse: The inverse of an implication $P \rightarrow Q$ is the statement $\neg P \rightarrow \neg Q$.
- It negates both the hypothesis and conclusion.
- Original: $P \rightarrow Q$"If P, then Q."
- Inverse: $\neg P \rightarrow \neg Q$ "If not P, then not Q."
- The inverse is also not necessarily logically equivalent to the original statement.
- Example:
- Original: "If it rains, then the ground is wet."
- Inverse: "If it does not rain, then the ground is not wet." (The ground could be wet for reasons other than rain, so this is also not logically equivalent.)

Converse, Inverse & Contrapositive in Propositional Logic

- 3. Contrapositive:
- The contrapositive of an implication $P \rightarrow Q$ is the statement $\neg Q \rightarrow \neg P$.
- It negates and reverses the hypothesis and conclusion.
- Original: $P \rightarrow Q$"If P, then Q."
- Contrapositive: $\neg Q \rightarrow \neg P$"If not Q, then not P."
- The contrapositive is logically equivalent to the original implication. If an implication is true, its contrapositive will also be true, and vice versa.
- Example:
- Original: "If it rains, then the ground is wet."
- Contrapositive: "If the ground is not wet, then it did not rain." (This is logically equivalent to the original statement.)

Statement	Form
Original Implication	$P \to Q$
Converse	$Q \to P$
Inverse	$\neg P \to \neg Q$
Contrapositive	$\neg Q \to \neg P$

Artificial Intelligence

Choose the correct choice(S) regarding the following proportional logic assertion S:

S: $((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$

(A) S is neither a tautology nor a contradiction.

(B) S is a tautology.

(C) S is a contradiction.

(D) The antecedent of S is logically equivalent to the consequent of S.

HW

Artificial Intelligence

Let p, q, r denote the statements "It is raining", "It is cold", and "It is pleasant", respectively. Then the statement "It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by

(A) $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$

(B) $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$

(C) $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$

(D) $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

H.W

Artificial Intelligence

Let p, q and r be propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is

(A) a tautology

(B) a contradiction

(C) always TRUE when p is FALSE

(D) always TRUE when q is TRUE

HW

Artificial Intelligence

The binary operator ^{1} is defined by the following truth table.

p	q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Which of the following is true about the binary operator $\neq$?

(A) Both commutative and associative

(B) Commutative but not associative

(C) Not commutative but associative

(D) Neither commutative nor associative

H.W

Artificial Intelligence

Let p, q, r, s represent the following propositions.

$p : x \in \{8, 9, 10, 11, 12\}$

q : x is a composite number

r : x is a perfect square

s : x is a prime number

The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____.

PW

Artificial Intelligence

Which one of the following Boolean expressions is NOT a tautology?

(A) $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$

(B) $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$

(C) $(a \wedge b \wedge c) \rightarrow (c \vee a)$

(D) $a \rightarrow (b \rightarrow a)$

H.W

Artificial Intelligence

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

I. $p \Rightarrow q$

II. $q \Rightarrow p$

III. $(\neg q) \vee p$

IV. $(\neg p) \vee q$

(A) I only

(B) I and IV only

(C) II only

(D) II and III only

H.W

Q9 The binary operation is defined as follows:

P	Q	P Q
T	T	T
T	F	T
F	T	F
F	T	T

Which one of the following is equivalent to P ∨ Q?

(A) ¬Q ¬P

(B) P ¬Q

(C) ¬P Q

(D) ¬P ¬Q

H.W

Q10 Consider the following statements:

P: Good mobile phones are not cheap

Q: Cheap mobile phones are not good

L: P implies Q

M: Q implies P

N: P is equivalent to Q

Which one of the following about L, M, and N is CORRECT?

(A) Only L is TRUE

(B) Only M is TRUE

(C) Only N is TRUE

(D) L, M and N are TRUE

PW

THANK - YOU